

Unit 5: Web, Email, Database Services, and Network Security

5.1 Apache, HTTPD, Email Operations (SMTP, Postfix, Dovecot)

1. Apache (HTTPD Web Server)

Apache HTTP Server (commonly called Apache or HTTPD) is a powerful, open-source web server software used to deliver web content over the Internet or an internal network. It processes client requests (from browsers) and sends back web pages, images, files, or application data using the HTTP (HyperText Transfer Protocol) and HTTPS (HTTP Secure) protocols.

Developed and maintained by the Apache Software Foundation, Apache is one of the most widely used web servers in the world.

Key Functions

- Hosts websites
- Handles client HTTP requests
- Supports virtual hosting
- Enables SSL/TLS encryption

Basic Configuration

Install:

```
sudo apt install apache2
```

Main configuration directory:

```
/etc/apache2/
```

Start/Restart:

```
sudo systemctl start apache2  
sudo systemctl restart apache2
```

Default Port:

- 80 (HTTP)
- 443 (HTTPS)

2. Email Operations (SMTP, Postfix, Dovecot)

Email servers use multiple protocols:

Protocol	Purpose
SMTP	Sending emails
POP3	Retrieving emails
IMAP	Managing emails on server

SMTP – Simple Mail Transfer Protocol

SMTP handles sending mail between servers.

Default Port:

- 25 (standard)
- 587 (submission)
- 465 (secure SMTP)

Postfix (Mail Transfer Agent)

Postfix is a widely used open-source Mail Transfer Agent (MTA) that is responsible for sending, receiving, and routing email messages between mail servers using the SMTP (Simple Mail Transfer Protocol).

It was developed as a secure and high-performance alternative to Sendmail and is commonly used on Linux/Unix servers for email infrastructure.

Install:

```
sudo apt install postfix
```

Main config file:

```
/etc/postfix/main.cf
```

Restart:

```
sudo systemctl restart postfix
```

Dovecot (Mail Delivery Agent)

Dovecot is an open-source Mail Delivery Agent (MDA) and IMAP/POP3 server that allows users to securely access and manage their email mailboxes.

While a Mail Transfer Agent like Postfix handles sending and routing emails, Dovecot is responsible for storing and providing access to those emails for end users.

Install:

```
sudo apt install dovecot-imapd dovecot-pop3d
```

Ports:

- 143 (IMAP)
- 993 (IMAPS secure)
- 110 (POP3)
- 995 (POP3S secure)

5.2 MySQL Administration

MySQL Database Server

MySQL is a relational database management system (RDBMS) used to store and manage structured data.

Key Tasks

- Create databases
- Manage users
- Backup & restore
- Secure database access

Install

```
'sudo apt install mysql-server
```

Secure installation:

```
sudo mysql_secure_installation
```

Basic Commands:

```
CREATE DATABASE mydb;  
CREATE USER 'user1'@'localhost' IDENTIFIED BY 'password';  
GRANT ALL PRIVILEGES ON mydb.* TO 'user1'@'localhost';
```

Default Port:

- 3306'

5.3 Cryptography, SSH, Firewall Configuration, ACLs

1. Cryptography (SSL/TLS)

SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are cryptographic protocols that encrypt communication between a client and a server over a network. They ensure that data transmitted over the Internet—such as passwords, emails, or web requests—remains private, secure, and tamper-proof.

TLS is the modern, more secure version of SSL and is widely used today.

Used in:

- HTTPS (Web)
- SMTPS (Email)
- Secure IMAP/POP
- SSH

Certificate tools:

- OpenSSL
- Let's Encrypt

2. SSH (Secure Shell)

SSH (Secure Shell) is a secure network protocol used to remotely access and manage servers over an encrypted connection. It allows administrators to log into a remote machine, execute commands, transfer files, and manage systems securely over unsecured networks like the Internet.

The most widely used implementation is OpenSSH, which provides secure remote login and encrypted communication between client and server.

Default Port:

- 22

Install:

```
sudo apt install openssh-server
```

Best Practices:

- Disable root login
- Use key-based authentication
- Change default port (optional)

3. Firewall Configuration

Common Linux firewall tools:

- UFW (Uncomplicated Firewall)
- iptables
- firewalld

Example using UFW:

```
sudo ufw allow 22  
sudo ufw allow 80  
sudo ufw enable
```

4. ACLs (Access Control Lists)

Access Control Lists (ACLs) are a mechanism in Linux/Unix systems that provide fine-grained permission control over files and directories, beyond the traditional `rwX` (read, write, execute) permissions for owner, group, and others.

ACLs allow specific permissions to be set for multiple users or groups on the same file or directory, offering more flexibility in access management.

Set ACL:

```
setfacl -m u:user1:rwX directory
```

View ACL:

```
getfacl directory
```

Anti-Spam Measures

Anti-spam measures are techniques and configurations implemented on email servers to prevent unwanted or malicious emails (spam) from reaching users' inboxes. Spam emails can carry phishing links, malware, or unwanted advertisements, so protecting the mail system is critical for security and productivity.

To secure email systems:

- ✓ Enable SPF
- ✓ Enable DKIM
- ✓ Enable DMARC
- ✓ Use spam filters like SpamAssassin
- ✓ Restrict open relay in Postfix

Troubleshooting Servers & Networks

Troubleshooting servers and networks is the systematic process of identifying, diagnosing, and resolving problems in server operations, network connectivity, and services. Effective troubleshooting ensures minimal downtime and maintains system security and performance.

Common Tools:

```
systemctl status service  
journalctl -xe  
netstat -tulnp  
ss -tuln  
ping  
traceroute
```

Check open ports:

```
ss -tulnp
```